

Lam Research Challenge Project Report

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1 Development of the Arena Circuit

We developed an interactive arena circuit that automates fluid delivery, displays status feedback through LCD indicators, and integrates signal-based control for gates, LEDs, and sensors. The setup coordinates with the Advanced Line Follower Robot (ALFR) as it moves through sequential gates, triggering fluid dispensing, visual indicators, and data collection at each stage.

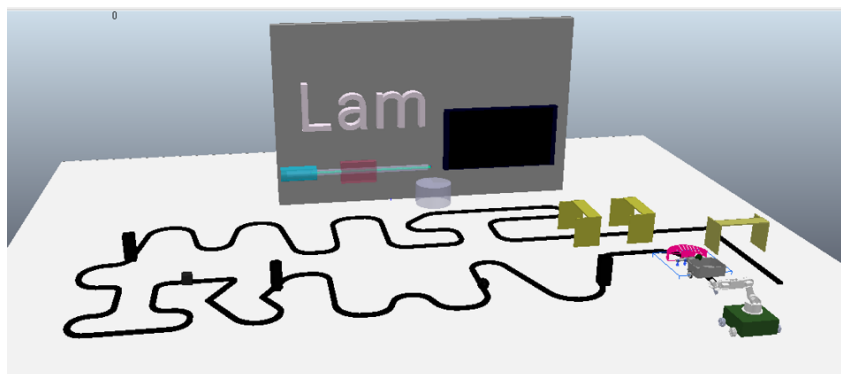


Figure 1: Arena

1.1 Overview of the Switch-Controlled Fluid Delivery Sequence

The arena is divided into three stages, each controlled via signals and sensor triggers:

Stage	Description	Action
Gate 1 (G1)	Fluid Delivery Trigger	Starts peristaltic pump to dispense 125 ml water
Gate 2 (G2)	LED Activation	Toggles LAM-shaped LED for indication
Final Stage	ALFR Parking & Load Measurement	Activates Lam-shaped LED and measures fluid weight

1.2 Water Parcel and Bowl Control (Gate 1 – Fluid Delivery)

1.2.1 Working Principle

1. The arena includes two predefined dummy-paths in CoppeliaSim.
2. When the robot crosses Gate 1:

- A Boolean flag `moveParcel` is set to true.
 - The parcel moves from a start dummy to an end dummy along the path, simulating fluid flow through the pipe.
 - Once the parcel completes its movement, the bowl starts moving toward the target to represent liquid dispensing completion.
3. The process visually represents a peristaltic pump transferring exactly 125 ml of liquid into a container.

1.3 LCD Display System

We provide a visual indication of the arena status:

- Task completed / failed
- Gate sequence status

1.3.1 Working Sequence

Initially, the screen is OFF. When the ALFR triggers the Gate 1 switch, water delivery starts. When the water is successfully dispensed, it shows “Simulation completed! Team: Mahagatbandhan”. This provides real-time simulation feedback.

1.4 Gate-Based Signal Control

- **Gate 1: Fluid Delivery Trigger:** The robot crosses the Gate 1 boundary. Once inside, it sets the signal `'trigger_water_release' = 1` & starts water movement (peristaltic simulation).
- **Gate 2: LED Activation (LAM Logo):** The robot crosses Gate 2. The signal is set to 1. This toggles the emissive property of the “LAM” logo mesh, creating a glowing LED effect.

1.5 Conclusion

The arena circuit was successfully developed to synchronize robotic movement, fluid transfer, and LED indication through coordinated signal-based control. The Lua scripts accurately replicate real-world behaviors such as pump activation, LCD status changes, and LED glow effects. This provides a complete automation ecosystem combining hardware logic and simulation, essential for demonstrating intelligent, event-driven fluid delivery.

2 Single Arm Robot (SARM) with Omni-Wheeled Platform

2.1 Platform

We have designed a completely new base platform from scratch, tailored to the competition arena and the mechanical requirements of the SARM. This custom design ensures:

- High maneuverability using four omni wheels for motion.
- Optimized internal space for safe component housing.
- Structural rigidity to carry the full SARM assembly without vibration or tilt.

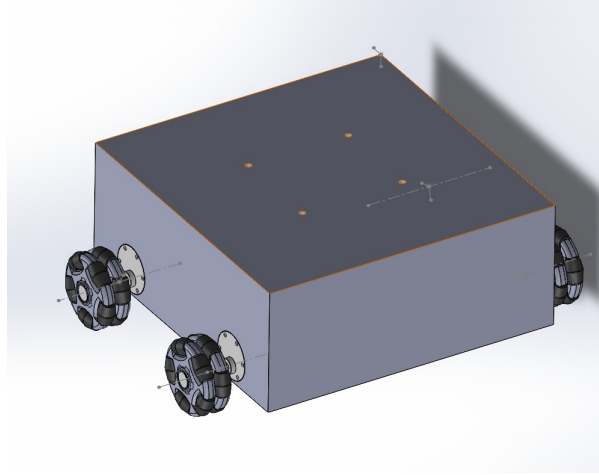


Figure 2: Custom Omni-Wheels Base

2.2 Arm Design and Modifications

The SARM was redesigned to better suit the platform and improve task efficiency. Key improvements include:

- **Gripper Modification:**

- Redesigned jaws to handle multiple object shapes.
- Prismatic Joints are designed with a desired max distance to pick different size objects.
- Maintains compactness to fit within arm dimensions and avoid interference during rotation.

2.3 Simulation Integration

After completing the CAD model, the assembly was exported to URDF (Unified Robot Description Format) for simulation in CoppeliaSim.

2.3.1 URDF & Coppelia Integration

- Each link and joint was parameterized (joint types, limits, masses).
- Collision models and inertia properties were defined for stable dynamics.
- Verified motion accuracy and gripper functionality through CoppeliaSim simulation.
- Ensured the center of mass lies within the platform for smooth omni movement.

The simulation confirmed stable movement, arm balance, and precise gripper control while handling various objects on the arena.

2.4 Highlights

- Fully custom omni-wheeled base designed to carry the robotic arm and electronics within a compact space.
- Modified arm & gripper capable of handling multiple object types while maintaining compactness.

- Integrated CAD $\rightarrow$ URDF $\rightarrow$ CoppeliaSim workflow, ensuring real-world accuracy in simulation.
- Balanced mechanical and control design, ensuring mobility, reach, and stability.

3 Advanced Line Follower Robot (ALFR)

Primary Function: The ALFR autonomously navigates a predefined line on the arena, intelligently detects and responds to obstacles using multiple sensors, and coordinates its behavior to optimize flow through the simulated environment.

3.1 Sensors in Use

3.1.1 Vision Sensors (5 units)

- **Names in Code:** Left2Sensor, LeftSensor, MiddleSensor, RightSensor, Right2Sensor.
- **Operational Logic:** Each sensor provides RGB data; a sensor is considered "seeing the line" if all RGB channels are very low (i.e., black line on a lighter arena). Motor velocities are dynamically adjusted in response to which sensors detect the line, allowing for sharp turns and smooth following. The UI actively toggles checkboxes representing each sensor's live state for debugging.

3.1.2 Proximity Sensor (1 unit)

- **Name in Code:** proximitySensor.
- **Operational Logic:** Continually reads the environment for obstacles. When a positive detection occurs, the robot stops both motors instantly. It remains stationary until no object is detected, at which point it resumes standard line-following motion.

3.2 Key Takeaways

- **Sensor-Driven Navigation:** Five vision sensors for robust line following.
- **Safety & Autonomy:** Proximity sensor ensures the robot halts at obstructions for autonomous "wait and go" behavior.
- **Simulation-Ready:** Lua code is CoppeliaSim compatible, with a debugging UI included.

4 Design & Development of Peristaltic Pump

The peristaltic pump simulation was done by moving a virtual Water Parcel and Water Bowl along pre-defined paths when triggered.

4.1 Working Principle

1. The arena includes two predefined dummy-paths in CoppeliaSim: `/Water_Path` and `/Water_Bowl_Path`.
2. The script uses `sim.getInt32Signal('trigger_water_release')` to start the process and `sim.setObjectInt32Param(...)` to toggle visibility of the fluid movement.

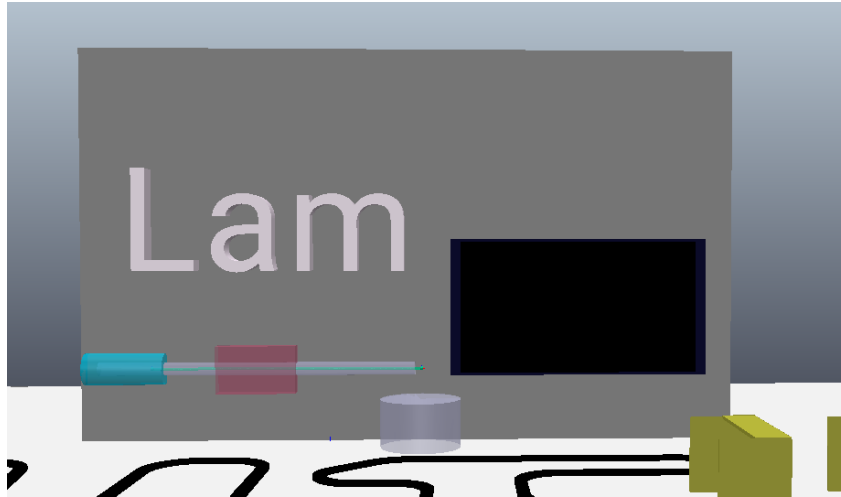


Figure 3: Peristaltic Pump Integration

4.2 Integration with Peristaltic Pump

In the real setup, this corresponds to:

- Pump ON signal → Dispense 125 ml.
- Pump OFF → When set volume is reached.

In simulation, this is visualized through water movement controlled by the Lua script. This represents the functioning of a peristaltic pump transferring exactly 125 ml of liquid.

5 3D Model – Custom Designs

5.1 Redesigned SARM with custom Gripper and Omni-Wheel Platform

We created a custom-designed 3D model that integrates the SARM with modified Grippers.

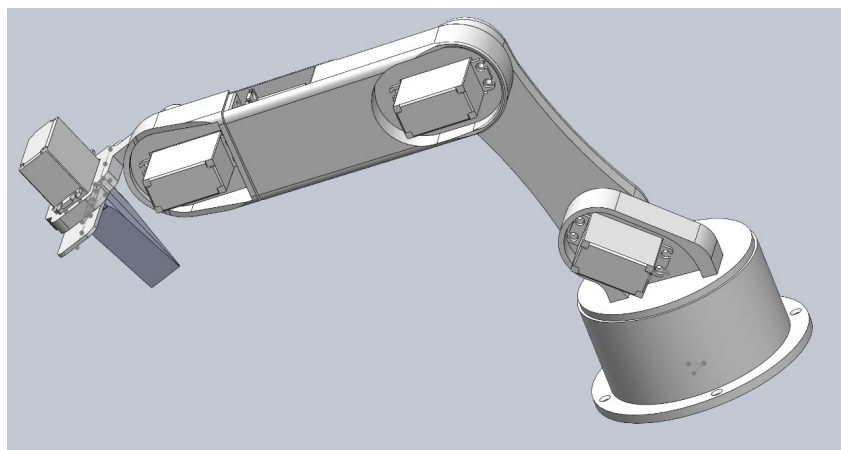


Figure 4: SARM with modified Grippers

5.1.1 Unique Features of our Bot

- **Original CAD Platform:** We designed our platform from scratch using SolidWorks, taking care of weight distribution for smooth omni-wheel control.

- **Custom Gripper Mechanism:** Features adjustable jaws for multi-shape object grasping and is optimized to fit the body while retaining strength.

5.2 Own Design for ALFR

We have designed a custom 3D model of the Advanced Line Follower Robot (ALFR) that fulfills all competition requirements while ensuring precision, modularity, and compactness.

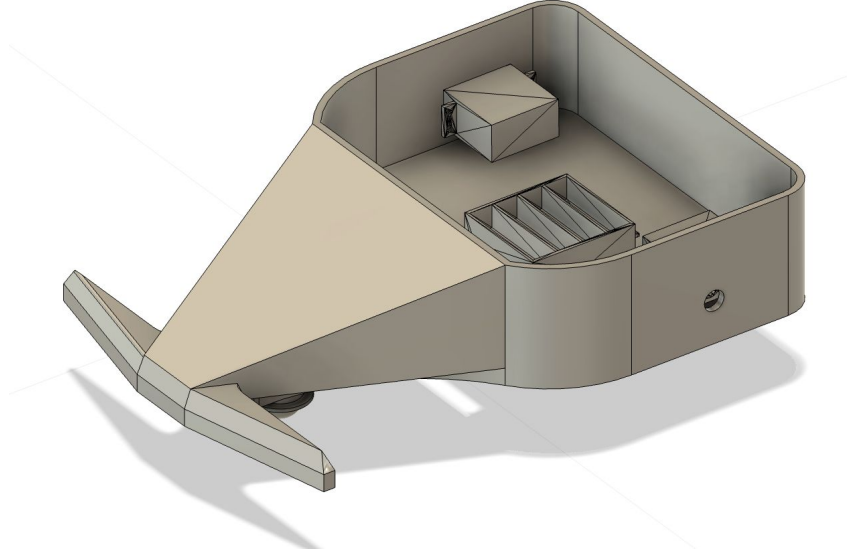


Figure 5: ALFR with 5 Vision Sensors

5.2.1 Design Overview

The custom ALFR design consists of a two-wheel drive system supported by a front castor wheel for balance. The chassis was designed to maintain stability, allow modular sensor mounting, and be lightweight for ease of assembly. The design was exported to CoppeliaSim as a URDF file for simulation.

5.2.2 Sensor Placement & Functionality

- **Vision Sensors:** Five vision sensors are mounted on a front strap at equal spacing to ensure continuous and accurate line detection.
- **Proximity Sensor:** A slot above the strap holds the proximity sensor for obstacle detection and avoidance.

6 Simulation Statistics

The total simulation time taken for the removal of 5 obstacles and the pump simulation is **2 minutes and 43 seconds**.

In CoppeliaSim, real-time depends on the computational load, physics calculations, and simulation step size. This means that the actual time observed while running the simulation (i.e., 8 minutes of recorded video) can be longer than the simulated task duration because the software may run slower than real-time depending on system performance.

Although the simulation video is 8 minutes long, the true simulation time representing the robots' task duration is 2 minutes and 43 seconds.

Acknowledgements

We would like to express our heartfelt gratitude to the LAM Research Challenge team for organizing this inspiring competition. Your guidance, support, and encouragement throughout the challenge have been invaluable. Participating in this event has been an enriching experience that pushed us to learn, innovate, and collaborate effectively as a team.

Thank you for providing us with this wonderful opportunity!



(a) Shreyans



(b) Dilshan



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(d) Sambhav

Figure 6: Our amazing team members from left to right.